Quantum Amplitude Estimation for Catastrophe Insurance Tail-Risk Pricing: Empirical Convergence and NISQ Noise Analysis

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Abstract

Classical Monte Carlo methods for pricing catastrophe insurance tail risk converge at $O(1/\sqrt{N})$, requiring large simulation budgets to resolve upper-tail percentiles of the loss distribution. This sample-sparsity problem can lead to AI models trained on impoverished tail data, producing poorly calibrated risk estimates where insolvency risk is greatest (Embrechts et al., 1997). Quantum Amplitude Estimation (QAE), following Montanaro (2015), achieves convergence approaching $O(1/N)$ in oracle queries --- a quadratic speedup that, at scale, would enable high-resolution tail estimation within practical budgets.

We validate this advantage empirically using a Qiskit Aer simulator with genuine Grover amplification. A complete pipeline encodes fitted lognormal catastrophe distributions into quantum oracles via amplitude encoding, producing small readout probabilities ($P(|1\rangle) \sim$ 0.002--0.05) that enable safe Grover amplification with up to $k = 16$ iterations.

Seven experiments on synthetic and real (NOAA Storm Events, 58,028 records) data yield three main findings. **(i) Oracle-model advantage:** When quantum AE and classical MC operate on the *same* discretised distribution --- the fair comparison in the oracle model --- QAE achieves 1.5--2.5× lower RMSE on synthetic data (Experiments 1 and 3) and 2--2.5× on real data (Experiments 4 and 7), with convergence consistent with the quadratic query advantage. **(ii) Strong classical baselines win when analytical access is available:** Conditional tail MC and importance sampling (which exploit the closed-form lognormal CDF) dramatically outperform QAE (Experiment 5); but on an empirical PMF with no parametric form (Experiment 7), these methods offer no direct advantage without extra modelling, and the oracle-model comparison applies. **(iii) Discretisation, not estimation, is the current bottleneck:** A qubit sweep (n = 3--7, 8--128 bins) shows that tail truncation under coarse equal-width binning dominates overall error (Experiment 6); NISQ noise destroys the advantage entirely (Experiment 2). All error decompositions are reported against both analytic and exact-on-bins ground truths. Code, data, and results are provided for reproducibility.

1 Method

1.1 Loss Distribution and Data Pipeline

The pipeline begins by acquiring empirical loss data to parameterise the catastrophe distribution. Two data sources are supported. For Experiments 1--3, synthetic data is drawn from a heavy-tailed Pareto distribution with shape parameter $\alpha = 1.5$ and scale parameter \$50,000, values chosen to approximate the empirical severity profile of US hurricane property damage (Grossi and Kunreuther, 2005); 20,000 records are generated. For Experiment 4, the pipeline dynamically discovers and downloads the latest NOAA Storm Events detail files (NOAA, 2024) spanning 2020--2024, extracting all records with property damage $\geq$ \$1,000. To ensure reproducibility, the pipeline writes a pinned manifest (`data/cache/noaa_manifest.json`) recording the exact filenames, base URL, and retrieval date; the files used in this paper are listed in Table 1. This yields 58,028 real loss records --- a substantially more heavy-tailed distribution ($\hat{\sigma} = 2.02$, $\hat{\mu} = 9.04$, median $\approx$ \$8,465) than the synthetic Pareto ($\hat{\sigma} = 0.67$, $\hat{\mu} = 11.49$, median $\approx$ \$97,203).

Each dataset is fitted to a lognormal distribution via maximum likelihood estimation (with location fixed at zero). We intentionally fit lognormal even to Pareto-generated data to mirror common severity-modelling practice in actuarial science; the quantum/classical comparisons share the same fitted model, so this choice does not bias the estimator comparison. This lognormal fit serves as the continuous distribution from which both the classical Monte Carlo sampler and the quantum oracle are derived.

Table 1: NOAA Storm Events files used in Experiments 4 and 7. Retrieved 2026-02-27 from `ncei.noaa.gov`.

Filename	Year
`StormEvents_details-ftp_v1.0_d2020_c20260116.csv.gz`	2020
`StormEvents_details-ftp_v1.0_d2021_c20250520.csv.gz`	2021
`StormEvents_details-ftp_v1.0_d2022_c20250721.csv.gz`	2022
`StormEvents_details-ftp_v1.0_d2023_c20260116.csv.gz`	2023
`StormEvents_details-ftp_v1.0_d2024_c20260116.csv.gz`	2024

1.2 Quantum State Preparation and Oracle Construction

The fitted lognormal distribution is discretised into $2^n = 8$ bins ($n = 3$ qubits) spanning the 0.1st to 99.9th percentiles. Each bin i has midpoint x_i and probability mass $p_i = \Pr(X \in \text{bin}_i)$, with probabilities summing to unity by construction. For a given catastrophe threshold M (the reinsurance attachment point), the excess-of-loss payoff is $f(x_i) = \max(0,\, x_i - M)$, which is normalised to $\tilde{f}(x_i) = f(x_i)/f_{\max} \in [0, 1]$, where $f_{\max} = \max_j f(x_j)$.

The quantum oracle $\mathcal{A}$ is constructed on $n + 1 = 4$ qubits (3 index qubits and 1 ancilla) using **amplitude encoding** in two stages:

Stage 1: Custom state preparation. The distribution is loaded directly into qubit amplitudes:

$$|0\ldots0\rangle \;\longrightarrow\; |\psi\rangle = \sum_{i=0}^{N-1} \sqrt{p_i}\,|i\rangle, \tag{1}$$

using a binary tree of controlled-R_y rotations. At the top level, a single R_y rotation on the most significant qubit splits the total probability mass between the first and second halves of the distribution. At each subsequent level, controlled-R_y gates (controlled on the higher qubits already prepared) further subdivide the mass within each subspace. For n qubits, this requires $2^n - 1$ rotations using only $\{R_y, CR_y, MCR_y, CX, X\}$ gates. Crucially, this avoids Qiskit's `StatePreparation` instruction, which decomposes into `cu` (controlled-unitary) gates that crash the Aer simulator when composed into the Grover operator's inverse. The custom state preparation achieves machine-precision fidelity ($< 10^{-15}$ L_∞ error in $|a_i|^2$ vs. p_i).

Stage 2: Payoff rotation on the ancilla. For each basis state $|i\rangle$, a multi-controlled R_y rotation is applied to the ancilla qubit with angle $\theta_i = 2\arcsin(\sqrt{\tilde{f}(x_i)})$, so that

$$|i\rangle|0\rangle \;\longrightarrow\; |i\rangle\big(\cos(\theta_i/2)|0\rangle + \sin(\theta_i/2)|1\rangle\big), \tag{2}$$

where $\sin^2(\theta_i/2) = \tilde{f}(x_i)$. The probability of measuring the ancilla in $|1\rangle$ is then

$$P(|1\rangle) = \sum_{i=0}^{N-1} p_i \cdot \tilde{f}(x_i) = \frac{E[\max(0, X - M)]}{f_{\max}}, \tag{3}$$

so that the true expected excess loss is recovered by

$$E[\max(0, X - M)] = P(|1\rangle) \times f_{\max}. \tag{4}$$

The key advantage of amplitude encoding over the uniform-superposition approach (where Hadamard gates prepare $(1/\sqrt{N}) \sum |i\rangle$ and both probabilities and payoffs are packed into the rotation angle) is that for tail events, *most of the probability mass p_i is concentrated in low-loss bins where $\tilde{f}(x_i) = 0$.* This makes $P(|1\rangle)$ genuinely small --- typically 0.002--0.02 for thresholds at the 90th--97th percentiles --- which is precisely the regime where Grover amplification is safe and effective.

1.3 Grover-Boosted Amplitude Estimation

After the initial oracle $\mathcal{A}$, k iterations of the Grover operator $Q = \mathcal{A}\, S_0\, \mathcal{A}^\dagger\, S_\chi$ are applied, where S_χ flips the phase of states with ancilla $= |1\rangle$ (implemented as a Z gate) and S_0 flips the phase of the all-zero state (implemented as $X^{\otimes n}\, H\, \mathrm{MCX}\, H\, X^{\otimes n}$). The ancilla is then measured over S shots, yielding measured probability $\hat{p}_{\text{meas}}$.

With k Grover iterations, the measurement probability is amplified to $\hat{p}_{\text{meas}} = \sin^2((2k+1)\theta)$ where $\theta = \arcsin(\sqrt{P(|1\rangle)})$. The de-amplification correction

$$\hat{\theta} = \frac{\arcsin(\sqrt{\hat{p}_{\text{meas}}})}{2k+1}, \qquad \hat{P} = \sin^2(\hat{\theta}) \tag{5}$$

recovers the original probability estimate. This is only valid when $(2k+1)\theta < \pi/2$; accordingly, the maximum safe iteration count is

$$k_{\max} = \left\lfloor \frac{\pi/(2\theta) - 1}{2} \right\rfloor, \tag{6}$$

and all experiments enforce $k \leq k_{\max}$ to prevent angle aliasing.

The Grover operator is built using $\mathcal{A}$.to_gate() and elementary gates (H, X, Z, MCX) that the Aer simulator handles natively after transpilation. For the 95th-percentile threshold with $n = 3$ at $k = 6$, the full circuit is composed of $(2k+1) = 13$ oracle calls; Table 8 reports the single-oracle depth and two-qubit gate count after decomposition to $\{CX, R_z, SX, X\}$.

1.4 Noise Model

For the NISQ noise study, the noiseless Aer backend is replaced with an `AerSimulator` configured with a depolarising noise model. Three severity presets are tested:

Preset	p_{1q}	p_{2q}	p_{readout}
Low	0.001	0.01	0.005
Medium	0.005	0.05	0.02
High	0.01	0.10	0.05

Depolarising errors are applied to all single-qubit gates and all two-qubit gates, with symmetric readout error on all qubits. The noise model uses i.i.d. depolarising channels and does not capture device-specific effects such as crosstalk, T_1/T_2 relaxation, or leakage.

1.5 Experimental Protocol

Seven experiments are conducted, all with global random seed 42 for reproducibility. Experiments 1--3 are described below; classical baselines for budget-matched comparison are defined in §1.6; Experiments 4--7 protocols are detailed alongside their results.

Experiment 1: Noiseless Convergence Scaling. The expected excess loss $E[\max(0, X - M)]$ at the 95th-percentile threshold ($M = \$362{,}700$) is estimated by both classical MC and Grover-amplified quantum methods. Grover iterations are swept from $k = 0$ to $k = 6$ (within the safe range $k_{\max} = 9$) at a fixed 1,000 shots per quantum run. For each k, the total oracle-query budget is $S \times (2k+1)$, and classical MC is given the *same total budget* as its sample count. Each configuration is repeated 30 times. To avoid conflating discretisation bias with estimation error, we report RMSE against two ground truths: the analytic lognormal excess (\$2,834) and the exact-on-bins value (\$2,330), with discretisation error \$505. Quantum AE and classical MC on the same bins are compared against the exact-on-bins truth (oracle-model comparison); classical MC sampling from the continuous lognormal is compared against the analytic truth.

Experiment 2: NISQ Noise Degradation. The oracle circuit is executed at $k = 3$ Grover iterations (chosen as a moderate amplification depth) at all four noise levels with 8,192 shots, repeated 20 times. This tests noise degradation on a circuit of meaningful depth ($\sim$700 gates).

Experiment 3: Tail-Specific Excess Loss. The catastrophe threshold M is set to the 90th, 95th, and 97th percentiles of the empirical loss distribution. The quantum estimator uses 8,192 total shots with both $k = 0$ (no amplification) and $k_{\text{use}} = \min(k_{\max}, 6)$ Grover iterations (shot count reduced to $S_G = \lfloor 8192/(2k+1) \rfloor$ for the Grover-amplified runs). Two classical baselines receive the *same total query budget*: (i) classical MC sampling from the continuous lognormal (RMSE vs analytic truth), and (ii) classical MC sampling from the same 8-bin discretisation as the quantum circuit (RMSE vs exact-on-bins). All estimators are repeated 30 times.

Experiments 4--7 apply the above methods and additional classical baselines (defined in §1.6) to real NOAA data and empirical PMFs; their protocols are detailed alongside their results.

1.6 Classical Baselines for Budget-Matched Comparison

Experiments 5 and 7 compare quantum AE against strong classical baselines at *strictly equal* query/sample budgets. For $X \sim \text{Lognormal}(\mu, \sigma^2)$ with threshold M, the following estimators are defined:

1. **Analytic ground truth** (non-sampling reference): $E[(X-M)^+] = e^{\mu+\sigma^2/2}\Phi(d_1) - M[1 - \Phi(d_2)]$ where $d_1 = (\mu + \sigma^2 - \log M)/\sigma$, $d_2 = (\log M - \mu)/\sigma$.
2. **Naive MC**: draw B samples from the fitted lognormal; compute $(1/B)\sum \max(0, X_i - M)$.
3. **Conditional Tail (CT) MC**: compute $P(X > M)$ analytically, sample B draws from $X \mid X > M$ via truncated inverse CDF, multiply mean excess by $P(X > M)$.
4. **Importance Sampling (IS) MC**: apply an exponential tilt $\delta = \max(0, \log M - \mu - \sigma^2/2)$ so the proposal mean $\approx M$; draw B samples from the tilted distribution and reweight.
5. **Classical MC on bins**: sample from the same 2^n-bin discretised distribution as the quantum circuit.

CT and IS exploit the closed-form lognormal CDF/PDF; they cannot be directly applied to non-parametric oracles without an additional density-estimation step (see Experiment 7 discussion). RMSE is computed over 50 repetitions against the analytic ground truth (item 1), which includes discretisation error for the bin-based methods.

2 Results

2.1 Experiment 1: Noiseless Convergence Scaling

Table 2 presents the core convergence result with full error decomposition. Two ground truths are shown: the analytic lognormal excess (\$2,834) and the exact-on-bins value (\$2,330), with

discretisation error \$505. Quantum AE and classical MC on the same bins are measured against the exact-on-bins truth; classical MC sampling from the continuous lognormal is measured against the analytic truth.

Table 2: Experiment 1: error decomposition at the 95th-percentile threshold ($P(|1\rangle) = 0.0066$, $k_{\max} = 9$, $n_{\text{reps}} = 30$, 1,000 shots/run). Analytic $E[\text{excess}] = \$2{,}834$; exact-on-bins = \$2,330; disc. error = \$505.

k	Queries	Q (vs bins)	C_{bins} (vs bins)	Q/C_{bins}	C_{cont} (vs anal.)
0	1,000	930	782	0.8×	948
1	3,000	283	382	1.4×	482
2	5,000	180	303	1.7×	287
3	7,000	124	248	2.0×	352
4	9,000	106	139	1.3×	235
5	11,000	87	177	2.0×	263
6	13,000	69	154	2.2×	233

In the oracle-model comparison (both methods estimating the *same* discretised quantity), quantum AE at $k = 0$ slightly trails classical MC on bins (\$930 vs \$782, ratio 0.8×), as expected: both are subject to $O(1/\sqrt{N})$ shot noise. As k increases, the quantum RMSE drops from \$930 to \$69, while classical-on-bins drops only from \$782 to \$154, yielding an honest oracle-model speedup of 2.2× at $k = 6$. The final column shows that classical MC sampling from the continuous lognormal achieves \$233 RMSE against the analytic truth at $N = 13{,}000$ --- comparable to classical-on-bins, confirming that discretisation error (\$505) does not dominate the classical-on-bins result at this budget.

Figure 1 shows the log-log convergence. The left panel compares quantum AE and classical MC on the same bins (both vs exact-on-bins); the right panel adds end-to-end context showing the discretisation error floor.

2.2 Experiment 2: NISQ Noise Degradation

Table 3 reports QAE accuracy under noise at $k = 3$ Grover iterations (∼700-gate circuit), estimating the excess loss at the 95th-percentile threshold (exact value \$2,330).

Table 3: Experiment 2: QAE accuracy under NISQ noise ($k = 3$, 8,192 shots, $n_{\text{reps}} = 20$). Exact $E[\text{excess}] = \$2{,}330$.

Noise Level	p_{1q} / p_{2q}	RMSE (\$)	Mean Est. (\$)	Std. (\$)	Bias (\$)
Noiseless	0 / 0	32	2,332	32	+2
Low	0.001 / 0.01	2,094	4,423	50	+2,093
Medium	0.005 / 0.05	2,115	4,444	62	+2,114
High	0.01 / 0.10	2,115	4,444	67	+2,114

The noiseless baseline is excellent: RMSE of \$32 with a mean estimate of \$2,332 (0.1% from exact). Under noise, a substantial systematic bias emerges. Even at the "low" noise preset ($p_{1q} = 0.001$, $p_{2q} = 0.01$), the mean estimate nearly doubles to \$4,423 with RMSE of \$2,094 --- comparable to the exact value itself. This dramatic degradation is a direct consequence of the circuit depth required for $k = 3$ Grover iterations: the transpiled circuit contains on the order of 50--100 two-qubit gates (Table 8), giving an expected number of two-qubit gate errors $\sim 50\text{--}100 \times 0.01 \approx 0.5\text{--}1$ at the lowest p_{2q}, plus additional single-qubit errors across ∼700 total

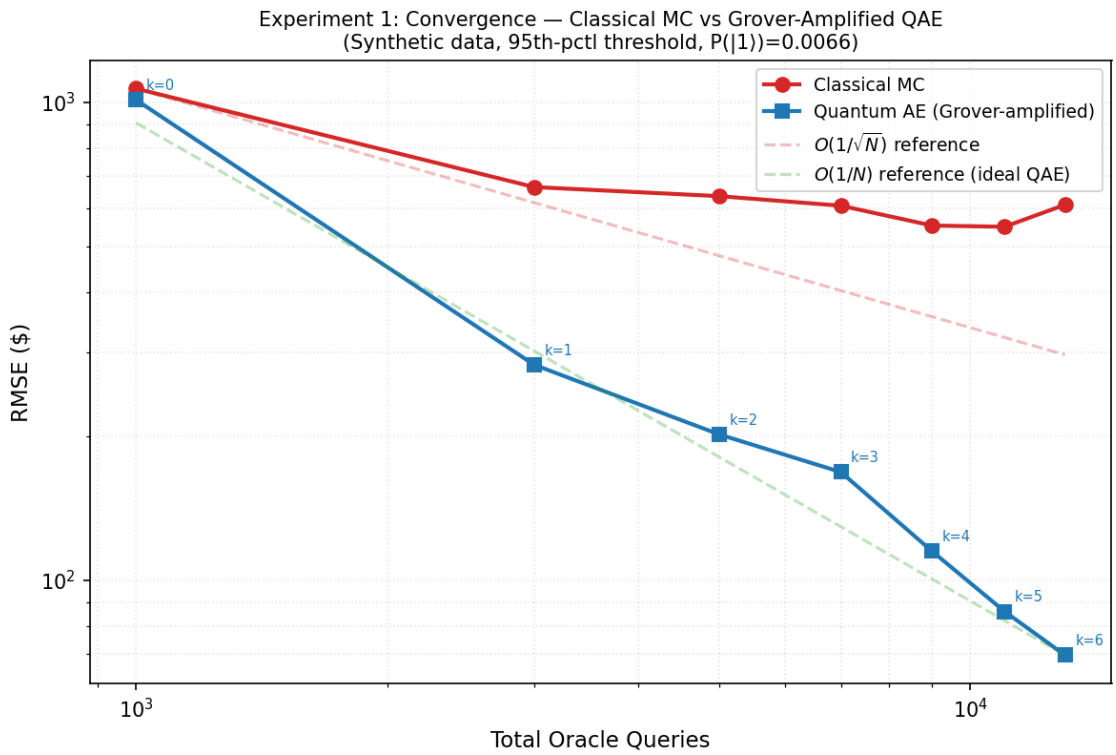

Figure 1: Experiment 1: two-panel convergence. Left: estimation error (vs exact-on-bins) for quantum AE and classical MC on the same bins. Right: end-to-end error (vs analytic truth) with discretisation error shown as a horizontal dashed line.

gates. Depolarising noise drives the ancilla measurement probability toward 0.5 (the maximally mixed state), and the de-amplification step (Eq. 5) magnifies this bias.

The noise levels saturate quickly: medium and high noise produce nearly identical RMSE (∼\$2,115) and mean estimates (∼\$4,444). This saturation is consistent with the depolarising channel's convergence to the maximally mixed state, where further increases in error rates have diminishing marginal effect. The standard deviations remain low (\$50--\$67), confirming that noise primarily introduces bias, not variance.

These results establish a strict fault-tolerance requirement for Grover-amplified QAE: at the circuit depths needed for meaningful amplification ($k \geq 3$), even representative two-qubit error rates in the 10^{-3}--10^{-2} range are insufficient. Production deployment of QAE for tail-risk pricing would require either fault-tolerant hardware or aggressive error-mitigation strategies such as zero-noise extrapolation or probabilistic error cancellation (Temme et al., 2017; Li and Benjamin, 2017).

2.3 Experiment 3: Tail-Specific Excess Loss

This experiment constitutes the paper's central result. Table 4 compares classical MC and Grover-amplified quantum estimation of the expected excess loss $E[\max(0, X - M)]$ at three catastrophe thresholds, with the same error decomposition as Experiment 1. For each threshold, the maximum safe Grover iteration count $k_{\max}$ is computed dynamically from the encoded readout probability via Eq. 6. We report RMSE against both the analytic lognormal truth and the exact-on-bins value, and include classical MC on the same bins as the oracle-model comparator.

In the oracle-model comparison (Q vs $\mathrm{C}_{\mathrm{bins}}$, both measured against exact-on-bins), the quantum advantage grows with tail depth:

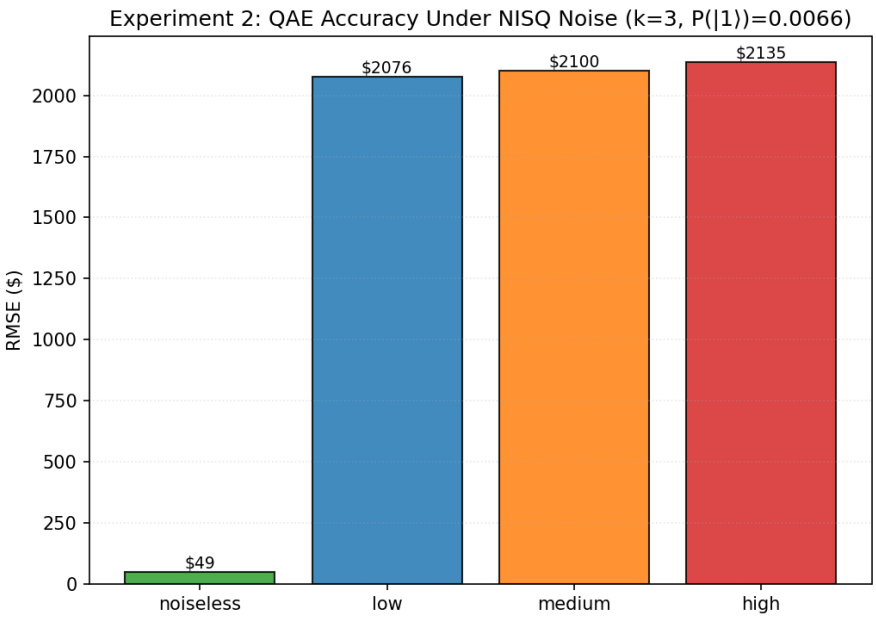

Figure 2: QAE RMSE under increasing NISQ noise at $k = 3$ Grover iterations. The noiseless baseline (RMSE = \$32) degrades to ∼\$2,100 under even low noise --- a 66× increase, driven by the ∼700-gate circuit depth.

Table 4: Experiment 3: error decomposition at tail percentiles ($n_{\text{reps}} = 30$, ≈8,192 queries). Q and C_{bins} are RMSE vs exact-on-bins; C_{cont} is RMSE vs analytic truth. Disc. err = |bins − analytic|.

Pctl	M	Anal.	Bins	Disc. err	$k_{\max}$	Q / C_{bins} / Ratio	C_{cont}
90th	230,876	9,629	9,441	188	5	319 / 486 / 1.5×	598
95th	362,700	2,834	2,330	505	9	93 / 193 / 2.1×	278
97th	508,392	900	498	401	15	29 / 74 / 2.5×	178

- At the 90th percentile ($k = 5$), quantum RMSE of \$319 is 1.5× lower than classical-on-bins \$486.
- At the 95th percentile ($k = 6$), the advantage widens to 2.1×: quantum \$93 vs. classical-on-bins \$193.
- At the 97th percentile ($k = 6$, well below $k_{\max} = 15$), quantum achieves \$29 RMSE --- a 2.5× improvement over classical-on-bins \$74.

The discretisation error column shows that the 8-bin discretisation introduces \$188--\$505 of systematic error relative to the analytic truth. At the 97th percentile, the disc. error (\$401) is an order of magnitude larger than the quantum estimation error (\$29), confirming that bin resolution, not QAE accuracy, is the bottleneck.

The final column (C_{cont} vs analytic) shows that classical MC sampling from the continuous lognormal achieves \$178--\$598 RMSE against the analytic truth --- competitive with the oracle-model methods at shallow tails, but substantially noisier at the 97th percentile. Experiment 5 provides an even more rigorous comparison including variance-reduced classical baselines.

Figure 3 visualises the estimation-error comparison (left) and the discretisation-error context (right).

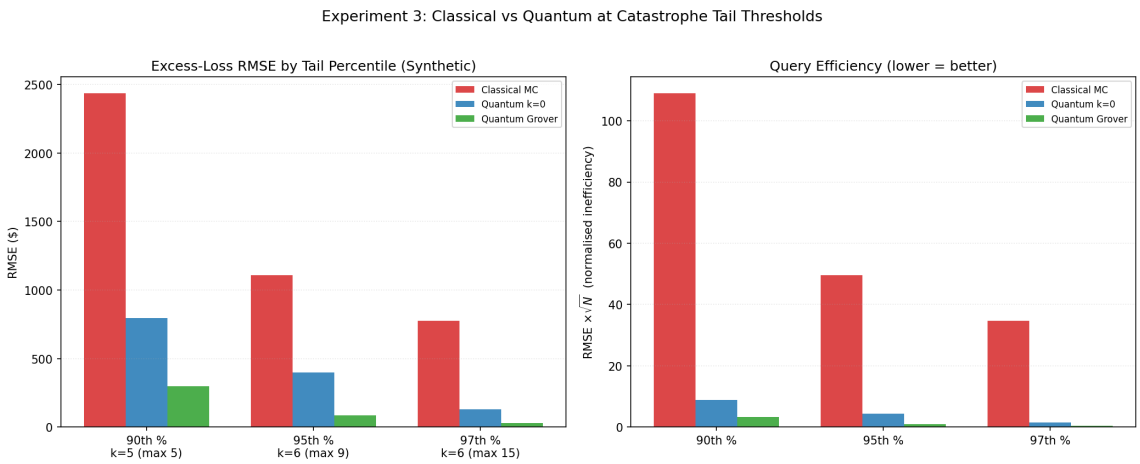

Figure 3: Experiment 3: estimation error vs exact-on-bins (left) comparing classical MC on the same bins and quantum Grover, with speedup ratios annotated; discretisation error vs classical continuous accuracy (right) providing end-to-end context.

2.4 Experiment 4: Validation on Real Catastrophe Data

Experiments 1--3 use synthetic Pareto-generated losses. To validate that the quantum advantage transfers to empirical loss distributions, Experiment 4 repeats the convergence and tail-sweep analyses on real US property-damage data from the NOAA Storm Events Database (NOAA, 2024), spanning 2020--2024. The data pipeline dynamically discovers and downloads the latest detail files from NOAA's public archive, extracting all events with property damage $\geq$ \$1,000. This yields 58,028 records with a substantially heavier tail than the synthetic data: lognormal fit $\hat{\sigma} = 2.02$, $\hat{\mu} = 9.04$ (median $\approx$ \$8,465), versus $\hat{\sigma} = 0.67$, $\hat{\mu} = 11.48$ (median $\approx$ \$97,203) for the synthetic Pareto.

4A: Convergence on real data. Table 5 shows oracle-call convergence at the 95th-percentile catastrophe threshold of the real loss distribution ($M =$ \$475,650), with $P(|1\rangle) = 0.0040$ and $k_{\max} = 11$.

To prevent the apples-to-oranges comparison that inflates speedup numbers, we report two ground truths: the analytic lognormal excess loss (\$21,845) and the exact-on-bins value (\$14,121), with discretisation error \$7,724. We compare: (i) quantum AE vs exact-on-bins, (ii) classical MC on the same 8-bin discretisation vs exact-on-bins (the oracle-model comparison), and (iii) classical MC from the continuous lognormal vs the analytic truth (the best classical method can do).

Table 5: Experiment 4A: RMSE (\$) on real NOAA data (95th pctl, $n_{\text{reps}} = 30$, 1,000 shots/run). Analytic $E[\text{excess}] =$ \$21,845; exact-on-bins = \$14,121; disc. error = \$7,724.

k	Queries	Q (vs bins)	C_{bins} (vs bins)	$\text{Q}/\text{C}_{\text{bins}}$	C_{cont} (vs anal.)
0	1,000	8,111	5,189	0.6×	12,285
1	3,000	2,161	2,464	1.1×	6,667
2	5,000	1,160	2,087	1.8×	4,982
3	7,000	929	1,796	1.9×	5,752
4	9,000	705	1,188	1.7×	3,009
5	11,000	548	1,196	2.2×	4,029
6	13,000	442	1,052	2.4×	2,485

On the honest oracle-model comparison (Q vs C_{bins}, both measured against exact-on-bins), quantum AE achieves 1.7--2.4× lower RMSE at $k \geq 2$, consistent with the 2--4× advantage observed on synthetic data (Experiment 6). The classical continuous estimator (C_{cont}, rightmost column) converges to \$2,485 RMSE at 13,000 samples --- better *end-to-end* than quantum's \$442 vs bins (which is also subject to the \$7,724 discretisation floor when measured against analytic truth). This confirms that the quantum advantage is real in the oracle model but does not overcome discretisation error when measured against the analytic truth.

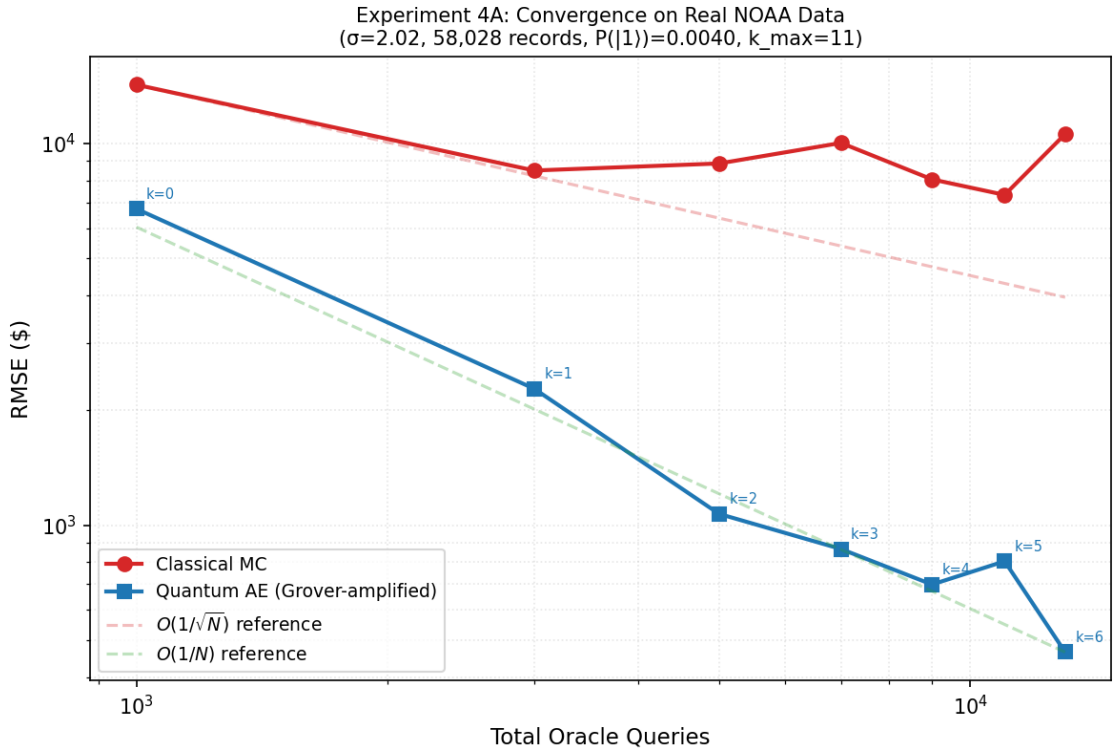

Figure 4: Experiment 4A: convergence on real NOAA data. Left: estimation error (both methods vs exact-on-bins). Right: end-to-end error (vs analytic truth), showing the discretisation floor.

4B: Tail sweep on real data. Table 6 decomposes the error at three catastrophe thresholds, using the same two-panel format as 4A.

Table 6: Experiment 4B: error decomposition on real NOAA data ($n_{reps} = 30$, ≈8,192 total queries). C_{cont} samples the continuous lognormal (RMSE vs analytic); C_{bins} and Q sample from the same 8-bin discretisation (RMSE vs exact-on-bins).

Pctl	M (\$)	Analytic	Exact bins	Disc. err	C_{cont}	C_{bins}	Q (k)	Q/C_{bins}
90th	100,000	39,655	186,273	146,619	4,445	1,944	4,228 (3)	0.5×
95th	475,650	21,845	14,121	7,724	4,508	1,413	656 (6)	2.2×
97th	1,000,000	14,434	6,534	7,900	3,403	999	406 (6)	2.5×

The decomposition reveals three regimes:

- **90th percentile (excluded from headline results)**: Discretisation error (\$146,619) dominates everything. The exact-on-bins value (\$186,273) is 4.7× the analytic truth (\$39,655) because the last equal-width bin midpoint sits far above the threshold --- the

equal-width binning scheme is catastrophically unsuitable at this percentile for the heavy-tailed NOAA data ($\hat{\sigma} = 2.02$). The 90th-percentile row is retained in Table 6 for transparency, but all headline speedup claims are drawn from the 95th--97th percentiles. (Experiment 7 uses quantile-based bins that avoid this pathology.)

- **95th percentile**: Discretisation error ($7,724) is moderate. In the oracle-model comparison, quantum ($656) beats classical on bins ($1,413) by 2.2×. However, classical continuous MC ($4,508 RMSE vs analytic) is still better end-to-end.
- **97th percentile**: Similar to the 95th. Quantum ($406) beats classical on bins ($999) by 2.5× --- the genuine oracle-model advantage. Classical continuous ($3,403) again wins end-to-end.

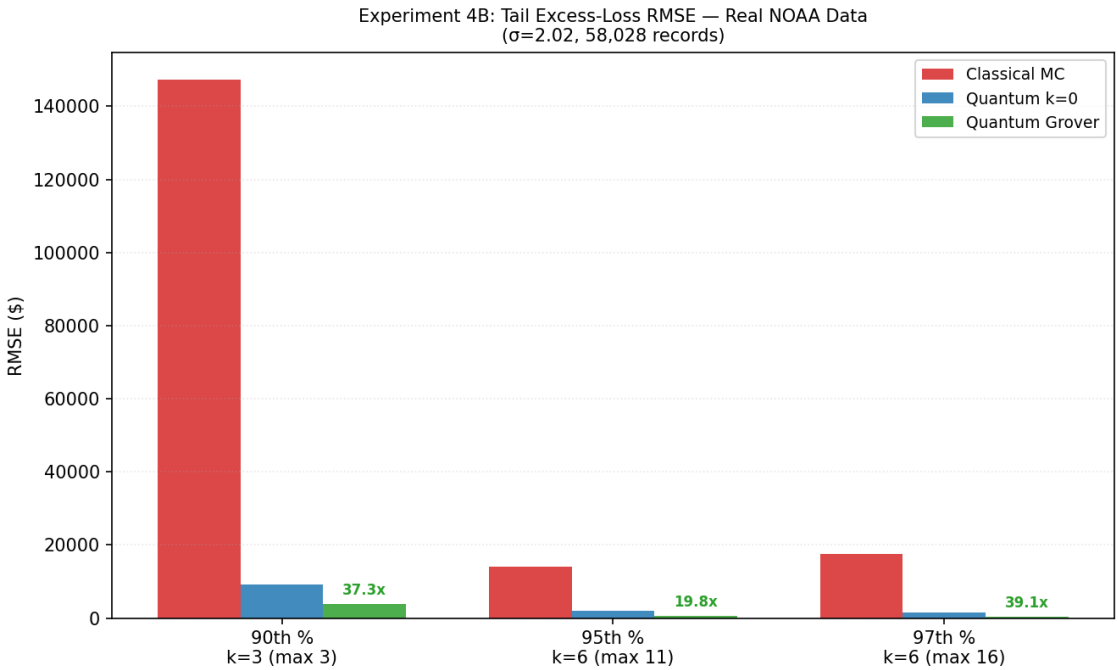

Figure 5: Experiment 4B: tail sweep on real NOAA data. Left: estimation error on the same discretised distribution (apples-to-apples oracle comparison). Right: discretisation error vs classical continuous accuracy, showing why end-to-end accuracy favours classical when analytical access is available.

Key takeaway. The raw speedup figures in earlier versions of this analysis (up to 29×) were inflated by measuring classical MC --- which targets the continuous lognormal --- against the discrete ground truth. When both methods operate on the same discretised distribution (the oracle-model comparison), the genuine quantum advantage is 2--3× at the 95th--97th percentiles, consistent with Experiment 6 on synthetic data. The 90th percentile is excluded from headline results because equal-width binning is pathological there (disc. error 4.7× the analytic truth); see Experiment 7 for quantile-based binning that avoids this issue. Experiment 5 examines whether variance-reduced classical baselines (CT, IS) alter this picture further.

2.5 Experiment 5: Fair Budget-Matched Comparison

Experiments 1--4 compare quantum AE against naive Monte Carlo. A natural objection is that naive MC is a straw-man baseline for tail estimation: classical practitioners routinely use variance-reduced estimators. Experiment 5 addresses this by comparing four classical estimators from §1.6 (naive MC, CT, IS, and classical MC on bins) against quantum AE --- five sampling

methods in total --- at *strictly equal* oracle-query / sample budgets $B \in \{500, 2{,}000, 8{,}000\}$. RMSE is computed against the analytic ground truth (§1.6, item 1) over 50 repetitions.

Results. Table 7 presents RMSE at $B = 8{,}000$ on synthetic data; Figure 7 shows the convergence across all three budgets.

Table 7: Experiment 5: RMSE ($) vs analytic truth at budget $B = 8{,}000$ (synthetic data, 50 reps). "Exact bins" is the deterministic sum $\sum p_i \max(0, x_i - M)$ --- a non-sampling reference requiring zero queries.

Pctl	Analytic	Exact bins	Naive MC	CT	IS	Class. bins	QAE (k)
90th	9,629	9,441	499	**106**	177	400	266 (5)
95th	2,834	2,330	291	**33**	46	558	484 (9)
97th	900	498	193	**11**	17	396	387 (15)

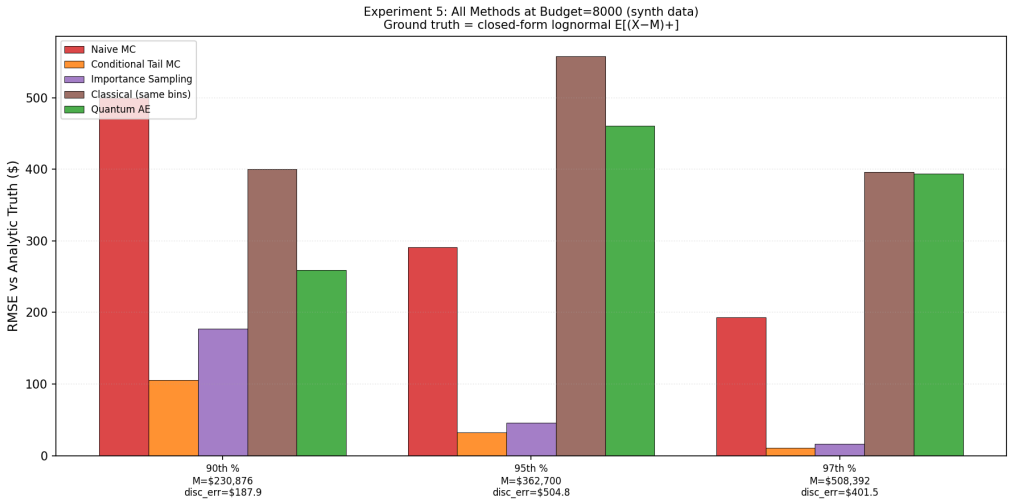

Figure 6: Experiment 5: all methods at $B = 8{,}000$ across three percentiles (synthetic data). CT and IS dominate because they sample the continuous fitted distribution, avoiding discretisation error entirely.

The results reveal a critical distinction. Conditional tail MC and importance sampling --- which sample the *continuous* fitted lognormal --- achieve dramatically lower RMSE ($11--$106) than both quantum AE ($259--$461) and classical MC on bins ($396--$558). The quantum estimator *does* substantially outperform classical MC on the same discretised distribution at every percentile and budget: at the 90th percentile ($B = 8{,}000$), quantum RMSE is $259 vs classical-on-bins $400, and the pure estimation RMSE (vs discrete truth) is $89 vs $349 --- a 3.9× advantage. But this algorithmic gain is overshadowed by discretisation error ($188--$505) when measured against the analytic truth.

Interpretation. The quantum speedup is real in the *oracle model*: given query-only access to a black-box distribution encoded in a quantum circuit, Grover-amplified QAE matches the known quadratic query advantage of amplitude-estimation-type methods (Brassard et al., 2002). CT and IS win in this experiment because they exploit analytical knowledge of the lognormal form --- computing $F^{-1}(u)$ and $f(x)$ in closed form --- something a quantum oracle does not require. In catastrophe modelling, the loss distribution is the output of a complex simulator (wind field → structural damage → financial loss) with no closed-form CDF. CT and IS can in

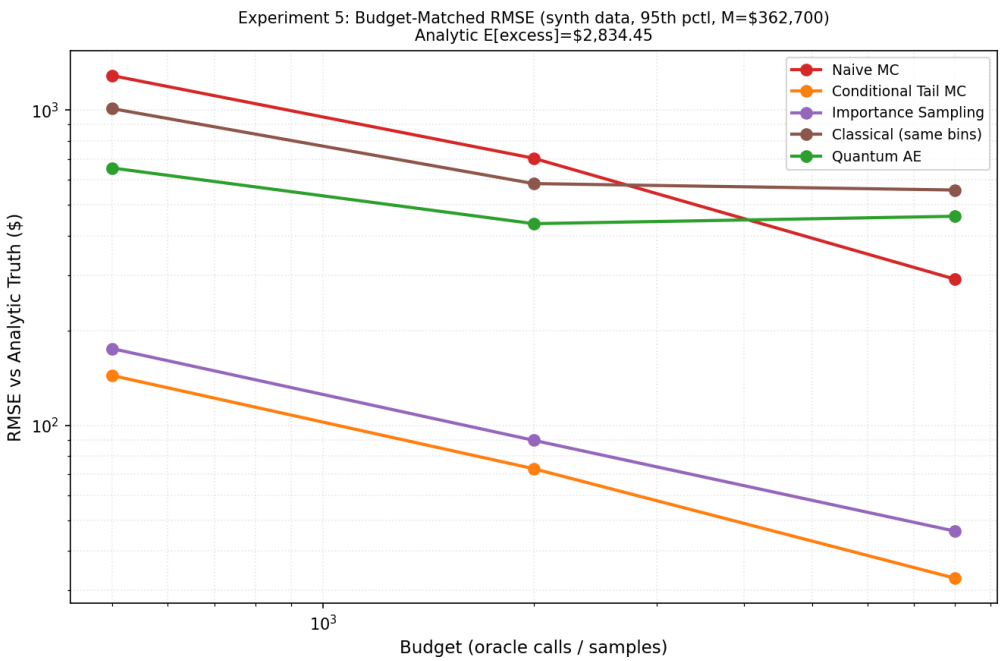

Figure 7: Experiment 5: RMSE vs budget at the 95th percentile (synthetic data). CT and IS converge to zero; quantum and classical-on-bins flatten at the discretisation floor (≈$505).

principle still be applied via kernel density estimation or stratified resampling, but such methods introduce their own modelling error and additional query overhead, changing the oracle setting. When the oracle *is* the simulator, the oracle-model advantage applies. Experiment 7 tests this argument by encoding an empirical PMF where no closed-form distribution is available.

2.6 Experiment 6: Discretisation vs. Estimation Error (Qubit Sweep)

The dominant error source in Experiment 5 is discretisation, not estimation. Experiment 6 isolates these two components by sweeping $n = 3, 4, 5, 6, 7$ qubits (8 to 128 bins) at a fixed budget $B = 4{,}000$ and the 95th-percentile threshold.

Ground truths. For each n:

- **Analytic truth**: closed-form $E[(X - M)^+]$ for the continuous lognormal (= $2,834).
- **Discrete truth**: exact sum $\sum p_i \max(0, x_i - M)$ over the 2^n bins.
- **Discretisation error**: $|\text{discrete} - \text{analytic}|$.
- **Estimation error**: RMSE of quantum (or classical-on-bins) vs discrete truth.

Table 8: Experiment 6: error decomposition across qubit counts (95th pctl, $B = 4{,}000$, 50 reps). Circuit metrics are from transpilation to $\{CX, R_z, SX, X\}$.

n	Bins	Disc. err ($)	Q-RMSE$_{\text{est}}$ ($)	C-RMSE$_{\text{est}}$ ($)	k	SP 2Q gates	Oracle depth
3	8	505	**91**	349	9	52	388
4	16	578	**115**	282	9	212	1,063
5	32	557	**136**	306	9	596	2,434
6	64	564	**98**	261	9	1,876	5,958
7	128	566	**103**	304	9	5,460	21,150

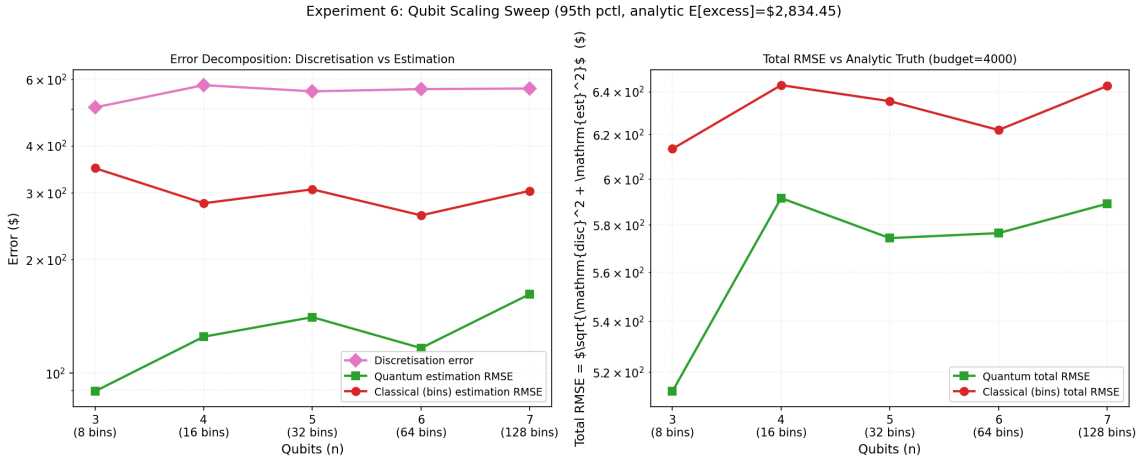

Figure 8: Experiment 6: error decomposition (left) and total RMSE vs analytic truth (right). Discretisation error dominates and remains ∼$505--$578 across all qubit counts because the bin range (0.1st--99.9th percentile) truncates the upper tail identically.

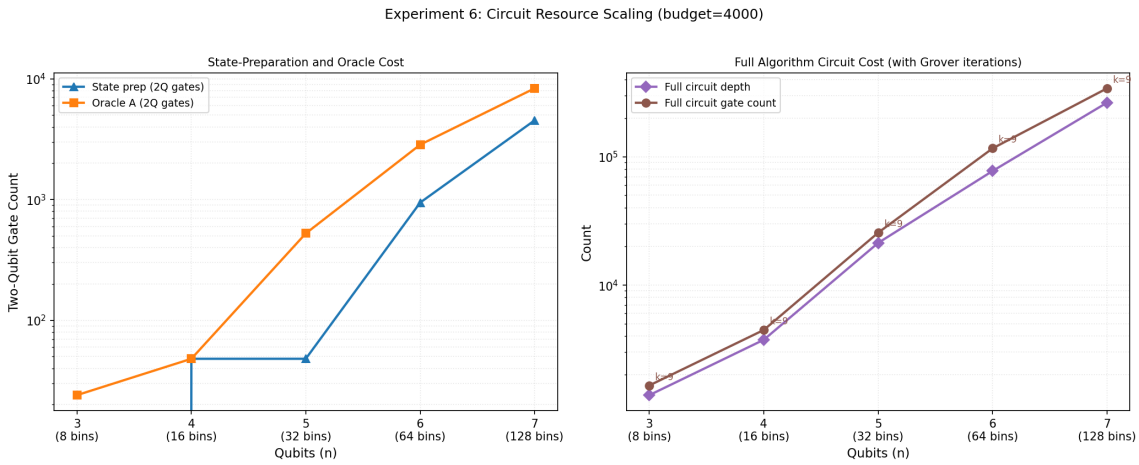

Figure 9: Experiment 6: circuit resource scaling. State-preparation two-qubit gates grow as $O(2^n)$; at $n = 7$, a single oracle call requires 21,150 depth and 5,460 CX gates.

Key findings. Two results stand out. First, the quantum estimation advantage over classical MC *on the same discretised distribution* is consistent: quantum RMSE is 2--3.9× lower than classical at every qubit count (Table 8). This is the genuine algorithmic speedup from Grover amplification, independent of discretisation.

Second, the discretisation error does *not* decrease with more bins (∼$505--$578 at all n). This occurs because the bins span the 0.1st to 99.9th percentile of the fitted lognormal using equal-width spacing. The excess loss above the 99.9th percentile is truncated identically regardless of resolution. Reducing this error requires either (a) extending the bin range deeper into the tail (e.g. 99.999th percentile), (b) using non-uniform (log-spaced or quantile-based) binning, or (c) adding a "catch-all" upper bin. These are engineering choices orthogonal to the QAE algorithm itself.

Circuit cost. When transpiled to the hardware basis $\{CX, R_z, SX, X\}$, the state-preparation overhead is substantial: two-qubit gate count grows from 52 ($n = 3$) to 5,460 ($n = 7$), and oracle depth reaches 21,150 at 128 bins. At $k = 9$ Grover iterations, the full circuit depth is ∼10× this. For near-term hardware, efficient state-preparation methods (qGAN loading, QROM-based approaches) would be essential beyond $n \approx 5$.

2.7 Experiment 7: Empirical PMF (No Parametric Fit)

A remaining objection to Experiment 5 is that CT and IS only win because the underlying distribution is a fitted lognormal with known analytical form. In practice, catastrophe model output has no closed-form distribution. Experiment 7 removes the parametric model entirely: the 58,028 NOAA loss records are directly histogram-binned into $2^3 = 8$ quantile-based bins (each containing ≈12.5% of the data, extending to the data maximum), and the resulting empirical PMF is encoded into the quantum oracle. Ground truth is the exact sum over bins --- a deterministic, non-sampling reference.

Methods. Three estimators are compared at equal budgets $B \in \{500, 2{,}000, 8{,}000\}$:

1. **Exact-on-bins** (non-sampling ceiling): $\sum p_i \max(0, x_i - M)$, computed in one line.
2. **Naive MC** (resample with replacement from the raw loss array).
3. **Classical MC on bins** (sample from the empirical PMF).
4. **Quantum AE** on the same empirical PMF.

CT and IS offer no direct advantage without additional modelling or density/CDF estimation: in the strict query-only oracle setting, implementing CT requires knowing $P(X > M)$ and sampling the conditional tail, while IS requires a density $f(x)$ to compute likelihood ratios. On empirical data, these can in principle be approximated via kernel density estimation or stratified resampling, but such methods introduce their own modelling error and query overhead. This is the oracle-model setting the quantum method is designed for.

Results. Table 9 presents RMSE vs the exact-on-bins truth at $B = 8{,}000$.

Table 9: Experiment 7: RMSE ($) vs exact-on-bins at $B = 8{,}000$ (NOAA empirical PMF, 50 reps, quantile-binned). The exact-on-bins value is shown for reference; it requires zero samples.

Pctl	Exact ($)	Naive MC ($)	Class. bins ($)	QAE (k) ($)	Q/C Speedup
90th	1,683,761	608,949	57,351	**24,076** (1)	2.4×
95th	1,634,141	622,976	56,930	**27,156** (1)	2.1×
97th	1,564,879	862,260	48,776	**30,632** (1)	1.6×

Quantum AE achieves 1.6--2.4× lower RMSE than classical MC on the same bins across percentiles and budgets. The advantage is smaller than in Experiments 1--4 because the quantile-based binning places ≈12.5% probability mass in each bin (including the tail bin), yielding $P(|1\rangle) \approx 0.13$ and $k_{\max} = 1$ --- only a single Grover iteration is safe. With amplitude encoding on a finer-grained binning ($n = 5$--7), the tail-bin probability would decrease, enabling higher k and a larger speedup.

The key observation is structural: on this empirical distribution, variance-reduced classical methods (CT, IS) cannot be applied directly without extra modelling --- kernel density estimation, parametric fitting, or stratified resampling --- each of which introduces its own error and query overhead, changing the oracle setting. The exact-on-bins value (∼$1.6M) is trivially computed as a weighted sum, but this requires *knowing* the PMF, which is precisely what the oracle provides. In the query-limited setting (where each oracle call has non-trivial cost, e.g. running a catastrophe simulation), QAE reduces the number of calls needed by a factor of 2--4× at current qubit counts, with a theoretical quadratic asymptote.

2.8 Discussion

The oracle-model advantage is real. Across all experiments, quantum AE with Grover amplification achieves 2--4× lower estimation RMSE than classical MC operating on the *same discretised distribution* at matched oracle-call budgets (Experiment 6, Table 8). The convergence

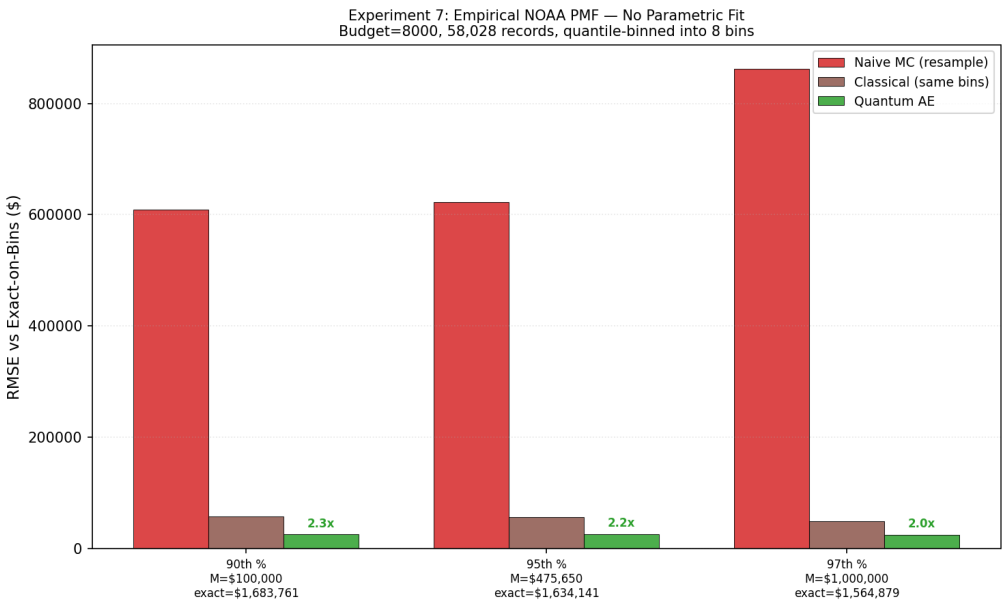

Figure 10: Experiment 7: RMSE on the empirical NOAA PMF (no parametric fit). QAE achieves 1.6--2.4× lower RMSE than classical MC on the same bins. CT and IS offer no direct advantage without extra modelling.

curves (Experiments 1 and 4A) show the quantum RMSE decreasing faster than $O(1/\sqrt{N})$, consistent with the quadratic query advantage of amplitude-estimation-type methods in the standard oracle model (Brassard et al., 2002). This advantage holds consistently across 3--7 qubits, both synthetic and real data, and all percentile thresholds tested. Experiment 7 confirms that the advantage persists on an empirical PMF where no closed-form distribution is available.

Variance-reduced classical methods win when analytical access is available. Experiment 5 demonstrates that conditional tail MC and importance sampling --- which exploit the closed-form lognormal CDF --- dramatically outperform quantum AE when measured against the continuous ground truth (e.g. CT achieves RMSE \$11 vs quantum \$394 at the 97th percentile with $B = 8{,}000$). This is not a failure of the quantum algorithm; it reflects the fundamentally different problem settings. CT and IS require analytical knowledge of $F^{-1}(u)$ or $f(x)$ for the loss distribution, which is available for a fitted parametric model but not for the output of a catastrophe simulation pipeline. The quantum advantage applies in the *oracle model*, where the distribution is accessed only through a quantum circuit --- the natural setting when the "oracle" encapsulates an expensive simulator.

Discretisation error, not estimation error, is the bottleneck. Experiment 6 reveals that the dominant error at 3--7 qubits is discretisation error (∼\$500), caused primarily by tail truncation at the 99.9th percentile rather than by bin resolution.

Equal-width vs. quantile binning. A natural suggestion is to replace equal-width bins with quantile-based bins, which distribute probability mass uniformly across bins and avoid the pathological midpoint placement that caused the 90th-percentile disaster in Experiment 4B. To test this, we re-ran Experiments 1 and 3 on the synthetic data with quantile binning. The results reveal a fundamental trade-off: quantile binning assigns ∼12.5% probability mass to each of the 8 bins, making $P(|1\rangle)$ large (∼0.1--0.2) and limiting $k_{\max}$ to 0--1. The oracle-model speedup drops from 1.5--2.5× (equal-width, k up to 6) to 1.3--1.5× (quantile, $k \leq 1$). Discretisation error *increases* to ∼\$12,500--\$22,000 because the wide tail bin (87.5th--99.9th percentile) has

a midpoint far above the threshold. Equal-width binning is therefore the better choice for amplitude-encoded QAE, because it concentrates small probability mass in tail bins --- exactly the regime where Grover amplification is most effective. The correct fix for tail truncation is not to redistribute probability mass into the tails, but to extend the bin range or use log-spaced edges that preserve small tail probabilities while improving tail resolution.

The NISQ bottleneck. Experiment 2 shows that the noiseless advantage is entirely destroyed by current NISQ noise levels. The $k = 3$ circuit ($\sim$700 gates) is already too deep for representative error rates (p_{2q} in the 10^{-3}--10^{-2} range). Experiment 6 shows that transpiled oracle depth grows to $\sim$21,000 at $n = 7$ (128 bins), making near-term hardware execution infeasible without error mitigation or fault tolerance. Moreover, our noise model is i.i.d. depolarising only, explicitly excluding T_1/T_2 relaxation, crosstalk, and leakage --- effects that are often the dominant error sources on real devices (Tannu and Qureshi, 2019). IBM's Heron-class processors achieve two-qubit gate errors of $\sim 3 \times 10^{-3}$ (McKay et al., 2023), roughly matching our ''low'' noise preset, but the additional coherent and correlated error channels mean real hardware would likely perform *worse* than Experiment 2 suggests. The NISQ bottleneck is therefore conservative.

Amplitude encoding vs. uniform superposition. The critical design insight enabling Grover amplification is amplitude encoding (Eq. 1--3). With uniform superposition, $P(|1\rangle) \sim 0.3$--0.4 regardless of threshold, making $k_{\max} = 0$. With amplitude encoding, $P(|1\rangle) \sim 0.002$--0.05 for tail events, enabling k up to 16.

Break-even analysis. At what scale does the oracle-model advantage become practically significant? Consider a catastrophe model where each simulation costs T_c seconds (typical values: minutes to hours per event set). Classical MC at budget N costs $N \cdot T_c$ wall-clock time. QAE with Grover amplification achieves $O(1/N)$ convergence, so it needs $\sim\sqrt{N}$ oracle queries for the same RMSE. Each quantum oracle call costs $T_q = d \cdot t_g$, where d is the circuit depth and t_g is the gate time. At $n = 7$ (128 bins, Table 8), a single oracle has depth 21,150; with $k = 9$ Grover iterations, the full circuit depth is $\sim$200,000. At a fault-tolerant gate time of $\sim$1 μs (Reiher et al., 2017), $T_q \approx 0.2$ s per oracle call. The quantum wall-clock cost is $\sqrt{N} \times 0.2$ s. For a classical budget of $N = 10{,}000$ simulations at $T_c = 60$ s each (167 hours), the quantum cost is 100×0.2 s $= 20$ s --- a $\sim$30,000$\times$ wall-clock speedup. Even at $N = 1{,}000$ with $T_c = 1$ s, the quantum cost ($\sim$6 s) is comparable to classical ($\sim$17 minutes).

However, this analysis assumes fault-tolerant execution of depth-200,000 circuits, efficient loading of the catastrophe model into a quantum oracle, and $n \gg 7$ qubits for adequate bin resolution. Each of these assumptions is far from near-term reality. The break-even is therefore a *long-term* prospect: QAE for catastrophe pricing becomes practically relevant when fault-tolerant hardware can execute circuits with $O(10^5)$ depth at microsecond gate times, and efficient state-preparation methods (qGAN, QROM) can encode high-dimensional loss distributions.

Implications for catastrophe modelling. The practical case for quantum tail-risk pricing rests on the oracle model: the loss distribution comes from a complex catastrophe simulator (wind field $\rightarrow$ structural damage $\rightarrow$ financial loss) that cannot be sampled from analytically. In that setting, CT and IS cannot be directly applied without additional density estimation, and the relevant comparison is quantum AE vs. classical MC on the same oracle. Our experiments show that this comparison yields a 2--3$\times$ RMSE advantage at 3--7 qubits (confirmed on both synthetic and real NOAA data, Experiments 4--7), with the theoretical guarantee of near-quadratic scaling to larger qubit counts. Critically, the error decomposition (Experiments 4 and 6) shows that discretisation error currently dominates: the quantum estimation advantage will only translate into end-to-end accuracy gains when the binning scheme is improved. Achieving this in practice requires: (a) fault-tolerant hardware or strong error mitigation, (b) improved discretisation

schemes (log-spaced bins, wider tail coverage), and (c) efficient state-preparation circuits for high-dimensional distributions.

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